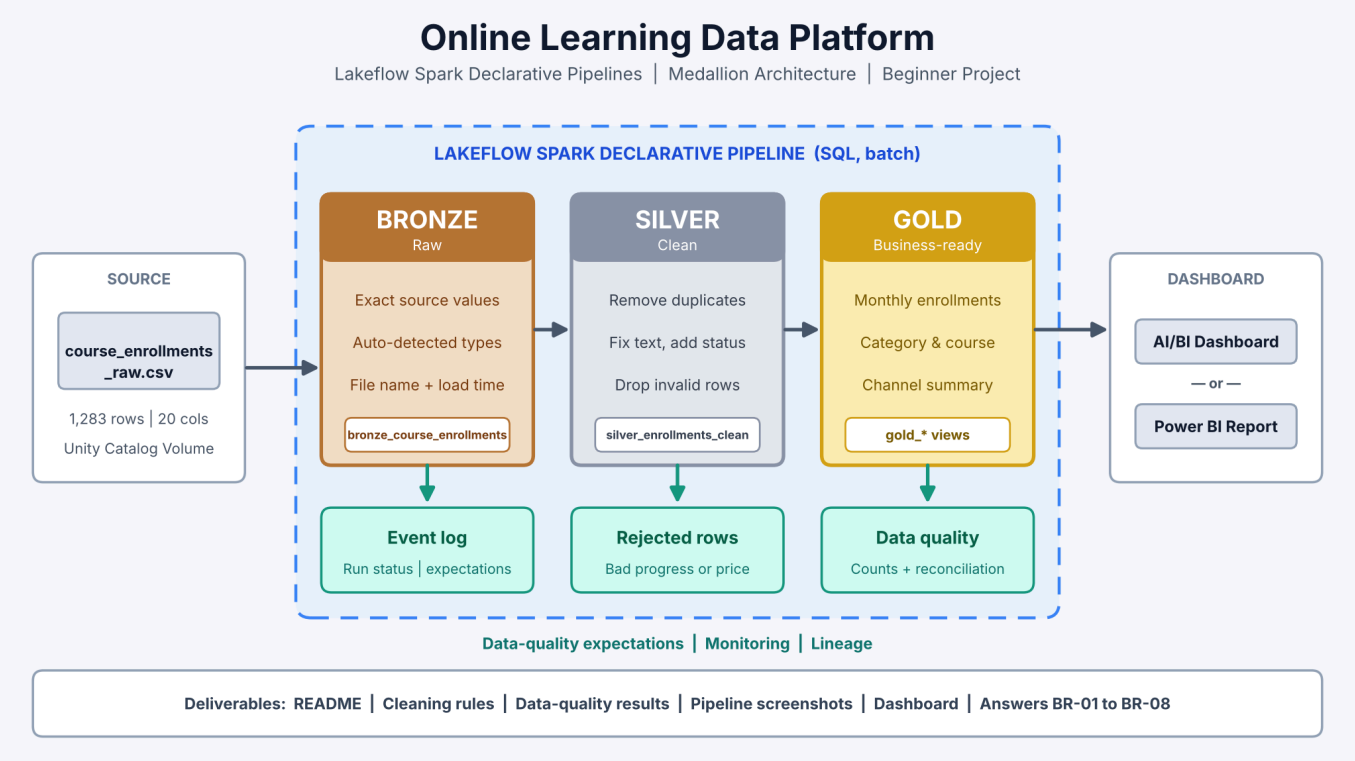
INNOMATICS RESEARCH LABS

Lakeflow Spark Declarative Pipelines

Online Learning Enrollment Analytics Platform



Project level: Beginner • Estimated time: 6–8 hours • SQL you need: SELECT, WHERE, GROUP BY, COUNT, SUM, AVG, DISTINCT, TRIM, INITCAP, LOWER, COALESCE, CASE WHEN, BETWEEN.

1. Executive Summary

An online learning platform sells video courses in data science, cloud, web development, design, business and marketing. It exports every course enrollment to a CSV file. Managers want simple, reliable reports on enrollments, revenue, course completion, learners and enrollment channels. The file has common problems: extra spaces, inconsistent capital letters, messy email addresses, blank values, duplicate enrollments, progress values above 100% and a few negative prices. You will build a Databricks Medallion Architecture that keeps the raw file, cleans it step by step, creates business-ready summary tables, and shows them in a dashboard.

Project goal: Transform a raw course enrollments CSV into trusted Gold-layer tables using Databricks SQL and Lakeflow Spark Declarative Pipelines, then build one dashboard on top of them.

2. Business Background

The platform offers 18 courses in six categories, taught by ten instructors at Beginner, Intermediate and Advanced level. Learners from ten Indian cities enroll through the Website, the Mobile App, a Referral or Social Media. Each enrollment has a list price, a discount, the price paid, the learner's progress (0–100%), hours watched and an optional course rating. A learner who reaches 100% progress gets a certificate. The same learner can enroll in several courses.

Current business challenges

- Category, course level, learner city and enrollment channel values have extra spaces and mixed capital letters.
- Some learner emails have capital letters or extra spaces, so the same learner looks like two different people.
- Some enrollments are missing the category, learner city, learner ID or price paid.
- Some enrollments appear twice, which makes enrollments and revenue look bigger than they are.
- A few enrollments show progress below 0% or above 100%, or a negative price paid.

3. Business Problem and Objectives

The platform has no automatic way to turn the raw enrollments file into numbers it can trust. This project builds a repeatable pipeline that keeps the original file, cleans the data, prepares summary tables and shows them in a dashboard.

Project objectives

- Load the enrollments CSV into Databricks from a Unity Catalog Volume.
- Keep the raw data in a Bronze table.
- Clean the data in a Silver table, add a learner status, and keep the bad enrollments in a separate rejected table.
- Create Gold summary tables for monthly enrollments, categories, courses and enrollment channels.
- Count the data problems, including how many learners were double-counted because of messy emails.
- Build one dashboard in Databricks AI/BI or Power BI.

4. Dataset Overview

The learning dataset course_enrollments_raw.csv is synthetic (made up for this project). It has 1,283 rows and 20 columns covering course enrollments in 2025. It looks like a real EdTech export but the learners and emails are not real.

Course Enrollments Dataset :- Click Here to Download Dataset

Dataset attribute	Description	Why it matters
Grain	One row = one learner enrolled in one course. enrollment_id is unique for each real enrollment.	Tells you how to count enrollments and find duplicates
Learner fields	learner_id, learner_email, learner_city, age_group	Supports learner counts and city reports

Dataset attribute	Description	Why it matters
Course fields	course_id, course_name, category, course_level, instructor_name	Supports category and course reports
Enrollment fields	enrollment_date, enrollment_channel, payment_method	Supports monthly and channel analysis
Money & progress fields	list_price, discount_amount, price_paid, progress_pct, hours_watched, course_rating, certificate_issued	Supports revenue and completion reports
Planted problems	Extra spaces, mixed case, messy emails, blanks, duplicates, progress outside 0–100, negative price	Gives you real cleaning practice

Column dictionary

Column	Example	Description
enrollment_id	EN500001	Unique ID of one enrollment
learner_id	LR7123	Learner ID (sometimes blank)
learner_email	arun.7123@gmail.com	Learner email (sometimes with capitals or spaces)
learner_city	Pune	Learner city (sometimes blank or messy)
age_group	25-34	18-24, 25-34, 35-44, 45+
course_id	C102	Course code
course_name	Machine Learning A to Z	Course title
category	Data Science	Course category (sometimes blank or messy)
course_level	Intermediate	Beginner, Intermediate, Advanced (sometimes messy)
instructor_name	Ravi Kumar	Course instructor
enrollment_date	2025-05-09	Date of enrollment
enrollment_channel	Mobile App	Website, Mobile App, Referral, Social Media (sometimes messy)
list_price	4499	Course price in ₹
discount_amount	899.80	Discount given
price_paid	3599.20	list_price − discount_amount (sometimes blank)
payment_method	UPI	UPI, Card, Net Banking, Wallet
progress_pct	65	Course progress from 0 to 100
hours_watched	31.4	Video hours watched
course_rating	4	Rating 1–5 (blank until 30% progress)
certificate_issued	Yes	Yes when progress is 100

Key business formulas

- price_paid = list_price − discount_amount
- learner_status = Completed (progress 100), In Progress (1–99) or Not Started (0)
- completion_rate_pct = Completed enrollments × 100 ÷ all enrollments
- unique learners = COUNT(DISTINCT learner_email) after cleaning the email

Important: enrollment_id should be unique. If the same enrollment_id appears more than once, the extra rows are duplicates. learner_email is NOT unique, because one learner can enroll in many courses.

5. Key Terms for Beginners

Term	Simple meaning
Unity Catalog Volume	A folder inside Databricks where you upload files such as CSVs.
Medallion Architecture	Organising data in three layers: Bronze (raw), Silver (clean), Gold (summaries).
Lakeflow Spark Declarative Pipelines	A Databricks feature where you write SQL that says what each table should contain; Databricks works out the order and runs it.
Streaming table	A table that reads new files incrementally. We use it for Bronze.
Materialized view	A saved query result that the pipeline keeps up to date. We use it for Silver and Gold.
Expectation	A data-quality rule, e.g. progress_pct BETWEEN 0 AND 100. With ON VIOLATION DROP ROW, bad rows are removed and counted.
INITCAP vs LOWER	INITCAP makes "data science" → "Data Science" (good for names). LOWER makes "Arun.7123@Gmail.com" → "arun.7123@gmail.com" (right for emails).
COUNT(DISTINCT ...)	Counts different values only once. Used to count unique learners.
Reconciliation	Checking that every input row is accounted for: kept, rejected or removed as a duplicate.

6. Proposed Architecture and Data Flow

Bronze keeps the raw enrollments. Silver cleans them and adds a learner status: good enrollments go to silver_enrollments_clean and bad enrollments go to silver_enrollments_rejected. Gold builds summary tables from the clean enrollments only, plus one data-quality table.

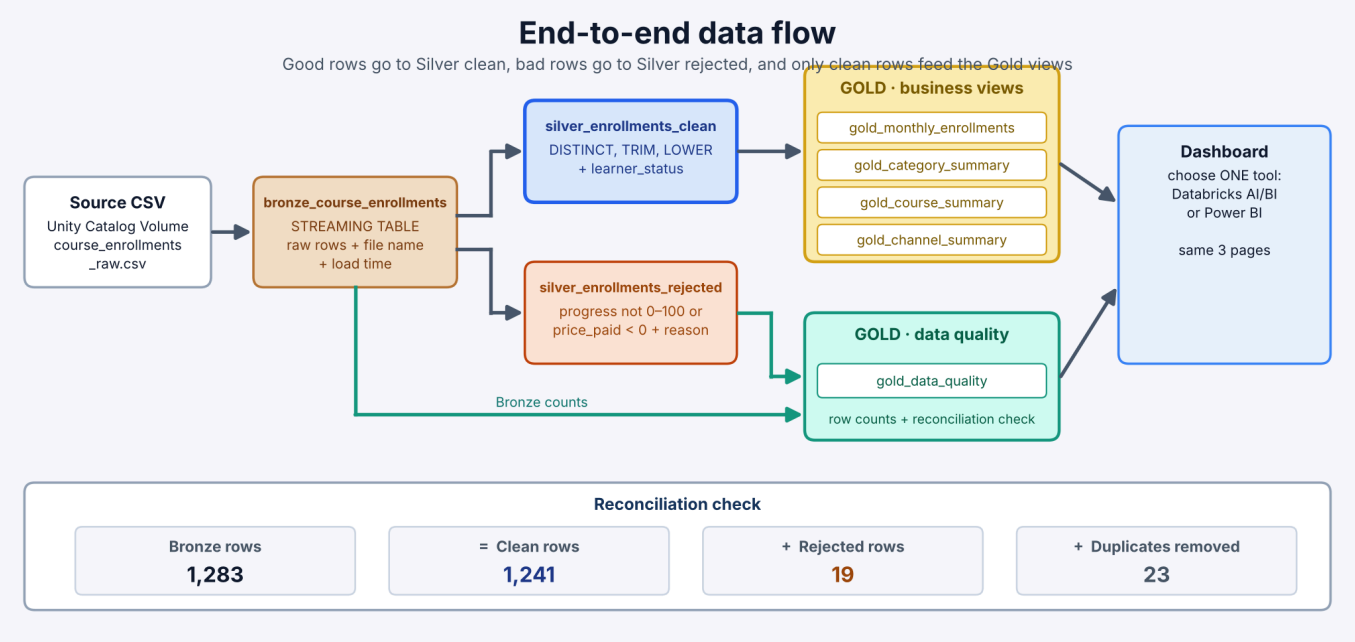


Figure 1 – End-to-end data flow and reconciliation check

7. Medallion Layer Responsibilities

Layer	Purpose	What you do	Result
Bronze	Keep the source data	Load the CSV as it is; add file name and load time	1 raw table
Silver	Clean the data	Remove duplicates, fix text and emails, fill blanks, add learner_status, drop impossible rows (and keep them in a rejected table)	1 clean table + 1 rejected table
Gold	Answer business questions	GROUP BY month, category, course and channel; calculate completion rate; count data problems	5 summary tables
Dashboard	Show the results	Build 3 pages in Databricks AI/BI or Power BI	1 dashboard

8. Business Requirements

ID	Requirement	Business expectation
BR-01	Monthly trends	Show enrollments, unique learners, revenue, discount, completions and hours watched for each month.
BR-02	Category performance	Compare enrollments, revenue, Completed / In Progress / Not Started counts and completion rate by category.
BR-03	Course performance	Compare enrollments, revenue, completion rate, average rating and hours watched by course.
BR-04	Channel analysis	Compare enrollments, revenue and completion rate by enrollment channel.
BR-05	Data-quality visibility	Show how many rows had each problem, and how many learners were double-counted before the email fix.

ID	Requirement	Business expectation
BR-06	Reconciliation	Prove that Bronze rows = clean rows + rejected rows + duplicates removed.
BR-07	Repeatability	Run everything as a Databricks pipeline that can be re-run with one click.
BR-08	Dashboard	Build one dashboard (Databricks AI/BI or Power BI) on the Gold tables.

9. Data Cleaning Rules

Silver cleaning is done in four simple steps. Each step uses one SQL idea.

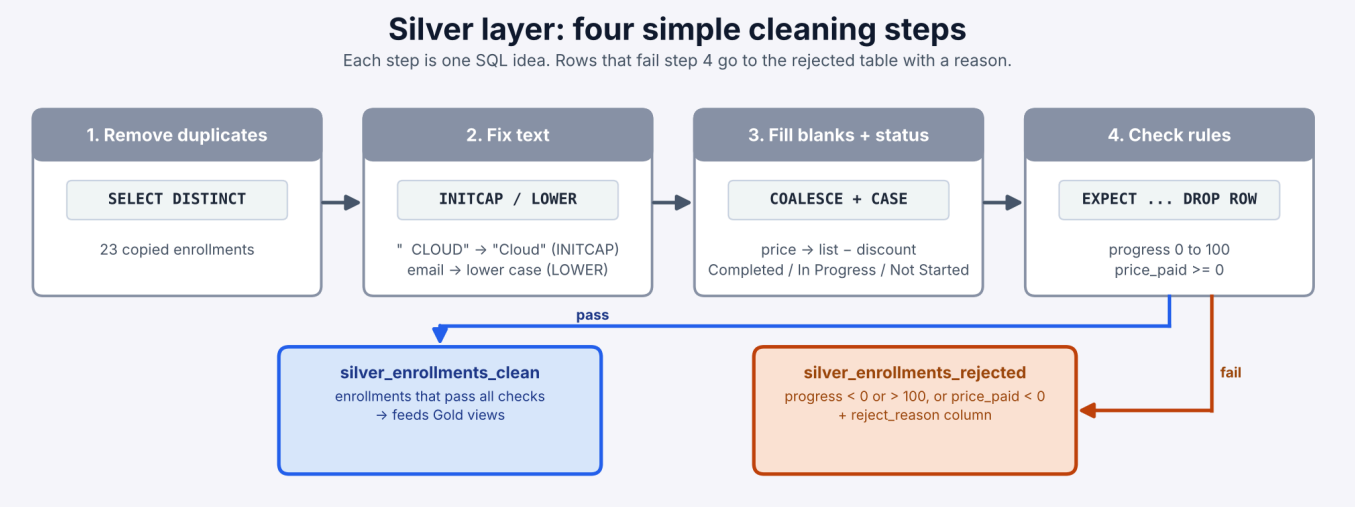


Figure 2 – The four Silver cleaning steps

Problem	Example	What to do	SQL hint
Duplicate enrollments	Same enrollment_id appears twice	Keep only one copy	SELECT DISTINCT
Extra spaces / mixed case	" cloud computing", "BEGINNER"	Remove spaces, use Title Case	INITCAP(TRIM(col))
Messy email	" Arun.7123@Gmail.com"	Remove spaces, use lower case	LOWER(TRIM(learner_email))
Blank category or city	category is empty	Replace with "Unknown"	COALESCE(col, 'Unknown')
Blank price paid	price_paid is empty	Calculate list_price − discount_amount	COALESCE(price_paid, list_price - discount_amount)
Blank rating or learner ID	course_rating is empty	Keep the row and leave it blank (AVG skips blanks)	No change
Impossible progress or price	progress_pct = 120 or -5; price_paid = -2999	Remove from clean table, keep in rejected table	EXPECT (progress_pct BETWEEN 0 AND 100) ON VIOLATION DROP ROW

New column: learner status

Add one new column in Silver with CASE WHEN, so Gold and the dashboard can group learners easily:

progress_pct	learner_status
100	Completed
1 to 99	In Progress
0	Not Started

Watch out: Progress of exactly 0 or exactly 100 is valid. Only values below 0 or above 100 are rejected. And never use INITCAP on emails: emails must be lower case.

10. Project Scope

Included

- One synthetic course enrollments CSV file.
- Batch loading from a Unity Catalog Volume.
- Bronze, Silver and Gold tables in one Lakeflow pipeline written in SQL.
- Simple data-quality checks and a reconciliation check.
- One dashboard in Databricks AI/BI or Power BI.

Not included

- Live video-watching events or real-time data.
- Refunds, subscriptions, quizzes or more than one source system.
- Real learner data and privacy controls (e.g. masking emails).
- Advanced data modelling (dimensions, SCD) and production alerting.

Scope note This is a learning project. Keep it simple, make it work end to end, and explain your choices.

11. Implementation Workflow

01 • Understand	Read the business questions and the 20 columns. Open the CSV and spot the problems yourself.
02 • Prepare	Create a catalog, schema and Volume. Upload the CSV.
03 • Ingest	Create the Bronze streaming table and check that 1,283 rows were loaded.
04 • Clean	Create the Silver clean and rejected tables using the four cleaning steps and the learner status.
05 • Curate	Create the four Gold business tables with GROUP BY.
06 • Validate	Create the Gold data-quality table and check the reconciliation.

07 • Publish & Visualize	Build one dashboard: Databricks AI/BI or Power BI.
08 • Explain	Present your pipeline, cleaning rules, results and dashboard.

Step-by-step guide with SQL hints

Use these starter snippets. Replace or complete the parts marked with "...".

Step 02 – Prepare the storage

```
CREATE CATALOG IF NOT EXISTS edtech_catalog;   -- or use a catalog you already have
CREATE SCHEMA  IF NOT EXISTS edtech_catalog.learning;
CREATE VOLUME  IF NOT EXISTS edtech_catalog.learning.raw_files;

-- Upload course_enrollments_raw.csv into the volume (Catalog Explorer → raw_files → Upload)
-- Then preview it:
SELECT * FROM read_files('/Volumes/edtech_catalog/learning/raw_files/',
                         format => 'csv', header => true) LIMIT 10;
```

Step 03 – Create the pipeline and the Bronze table

Create a new ETL pipeline (Lakeflow Spark Declarative Pipelines), set the default catalog to edtech_catalog and schema to learning, and add a SQL file with:

```
CREATE OR REFRESH STREAMING TABLE bronze_course_enrollments AS
SELECT *,
       _metadata.file_name AS source_file,
       current_timestamp() AS loaded_at
FROM STREAM read_files('/Volumes/edtech_catalog/learning/raw_files/',
                       format => 'csv', header => true);
```

Step 04 – Silver clean and rejected tables

```
CREATE OR REFRESH MATERIALIZED VIEW silver_enrollments_clean (
  CONSTRAINT valid_progress EXPECT (progress_pct BETWEEN 0 AND 100) ON VIOLATION DROP ROW,
  CONSTRAINT valid_price    EXPECT (price_paid >= 0)                ON VIOLATION DROP ROW
) AS
WITH cleaned AS (
  SELECT DISTINCT
    enrollment_id, course_name, enrollment_date, progress_pct, discount_amount,
    LOWER(TRIM(learner_email))                          AS learner_email,
    COALESCE(INITCAP(TRIM(category)), 'Unknown')        AS category,
    INITCAP(TRIM(course_level))                         AS course_level,
    COALESCE(price_paid, list_price - discount_amount)  AS price_paid,
    ...   -- add the other columns; clean learner_city and enrollment_channel too
  FROM bronze_course_enrollments
)
SELECT *,
  CASE WHEN progress_pct = 100 THEN 'Completed'
       WHEN progress_pct > 0   THEN 'In Progress'
       ELSE 'Not Started' END AS learner_status
FROM cleaned;

CREATE OR REFRESH MATERIALIZED VIEW silver_enrollments_rejected AS
SELECT *,
  CASE WHEN progress_pct < 0 OR progress_pct > 100 THEN 'Progress is outside 0 to 100'
       ELSE 'Price paid is negative' END AS reject_reason
FROM bronze_course_enrollments
WHERE progress_pct < 0 OR progress_pct > 100 OR price_paid < 0;
```

Step 05 – Gold business tables

```
CREATE OR REFRESH MATERIALIZED VIEW gold_monthly_enrollments AS
SELECT DATE_TRUNC('MONTH', enrollment_date)                    AS enrollment_month,
       COUNT(*)                                                AS enrollments,
       COUNT(DISTINCT learner_email)                           AS unique_learners,
       SUM(price_paid)                                         AS revenue,
       SUM(CASE WHEN learner_status = 'Completed' THEN 1 ELSE 0 END) AS completions,
       SUM(hours_watched)                                      AS hours_watched
FROM silver_enrollments_clean
GROUP BY DATE_TRUNC('MONTH', enrollment_date);
```

Build the other three Gold tables with the same pattern:

Gold table	GROUP BY	Columns to calculate
gold_category_summary	category	enrollments, revenue, completed, in_progress, not_started (SUM with CASE WHEN), completion_rate_pct, avg_progress_pct
gold_course_summary	category, course_name, course_level, instructor_name	enrollments, revenue, completion_rate_pct, avg_rating (AVG of course_rating), hours_watched
gold_channel_summary	enrollment_channel	enrollments, revenue, completion_rate_pct (Completed × 100.0 ÷ COUNT(*))

Step 06 – Data quality and reconciliation

```
CREATE OR REFRESH MATERIALIZED VIEW gold_data_quality AS
SELECT COUNT(*)                                                  AS bronze_rows,
       COUNT(*) - COUNT(DISTINCT enrollment_id)                  AS duplicate_rows,
       SUM(CASE WHEN progress_pct < 0 OR progress_pct > 100 THEN 1 ELSE 0 END) AS invalid_progress_rows,
       COUNT(DISTINCT learner_email)                             AS learners_before_email_fix,
       COUNT(DISTINCT LOWER(TRIM(learner_email)))                AS learners_after_email_fix,
       ...   -- add negative price, missing category, missing city, missing price
       (SELECT COUNT(*) FROM silver_enrollments_clean)           AS clean_rows,
       (SELECT COUNT(*) FROM silver_enrollments_rejected)        AS rejected_rows
FROM bronze_course_enrollments;
```

Then run the pipeline (Start), open the pipeline graph, and check: **bronze_rows = clean_rows + rejected_rows + duplicate_rows**. Also compare learners_before_email_fix with learners_after_email_fix.

12. Dashboard (Choose Databricks AI/BI or Power BI)

The dashboard is the last stage of the project. It reads only the Gold tables. Build it in **one** tool only: Option A or Option B. Both give the same three pages.

Choose one option: If you build the dashboard in Power BI, skip Option A. If you build it as a Databricks AI/BI dashboard, skip Option B.

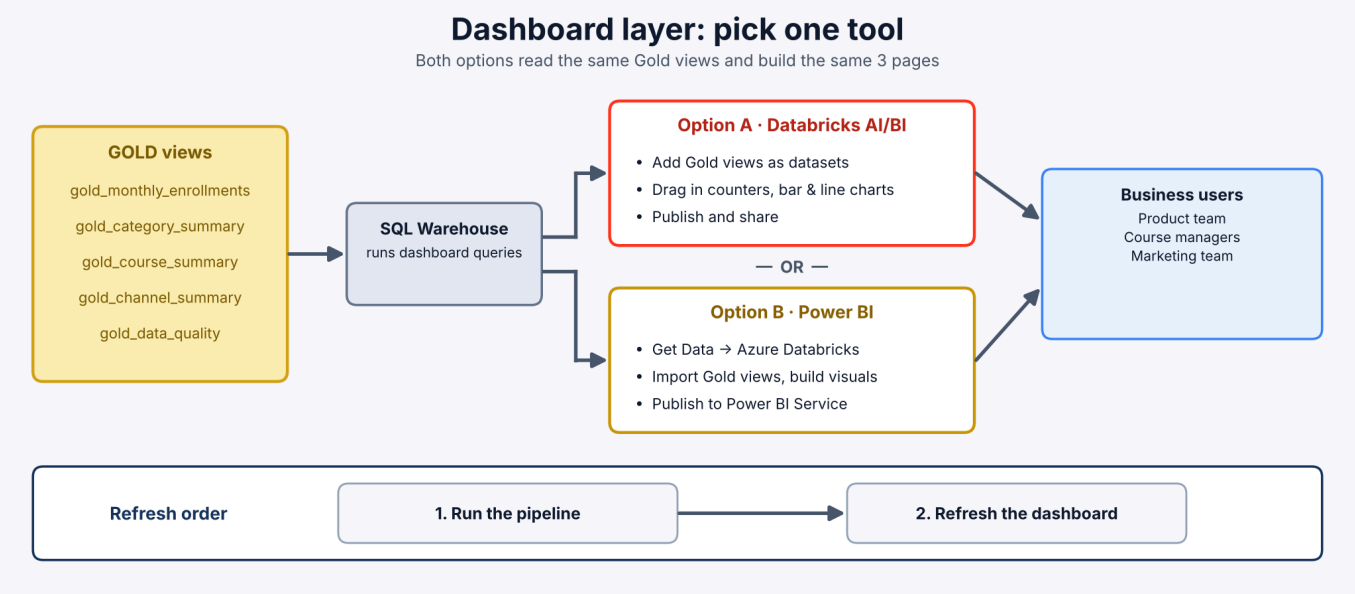


Figure 3 – Dashboard layer: one tool, same Gold tables

Dashboard pages (same for both options)

Page	Visuals	Gold table
1. Overview	KPI cards: Enrollments, Revenue, Completions, Hours Watched; line chart of revenue by month; bar chart of revenue by category; bar chart of learners by status	gold_monthly_enrollments, gold_category_summary
2. Courses & Channels	Table of top 10 courses with revenue, completion rate and rating; bar chart of completion rate by category; bar chart of enrollments and revenue by channel	gold_course_summary, gold_category_summary, gold_channel_summary
3. Data Quality	Cards: Bronze rows, clean rows, rejected rows, duplicates; learners before vs after email fix; reconciliation passed (Yes/No); bar chart of problem counts	gold_data_quality



Figure 4 – Example Overview page (numbers from the reference solution)

Option A – Databricks AI/BI Dashboard

- **Create:** in Databricks choose New → Dashboard.
- **Add data:** on the Data tab, add each Gold table as a dataset (pick the table or write SELECT * FROM ...).
- **Add visuals:** on the canvas, add Counter widgets for the KPIs (sum of enrollments, revenue, completions, hours_watched), a Line chart for monthly revenue and Bar charts for category, learner status, course and channel.
- **Add filters:** add a filter for month, category or course level where those columns exist.
- **Publish:** click Publish and share the dashboard with your team. Optionally add a refresh schedule.

Option B – Power BI Report

- **Connect:** in Power BI Desktop choose Get Data → Azure Databricks. Copy the Server hostname and HTTP path from your SQL warehouse (Connection details tab) and sign in.
- **Load data:** open edtech_catalog → learning, tick the five gold_* tables, and choose Import.
- **Add visuals:** use Card visuals for the KPIs (Sum of enrollments, revenue, completions, hours_watched), a Line chart for monthly revenue and Bar charts for category, learner status, course and channel.
- **Percentages and unique counts:** show completion_rate_pct, avg_rating and unique_learners per row in tables or bar charts. Do not add them up: a learner who enrolls in two months would be counted twice.
- **Add filters:** add Slicers for month, category or course level on the pages where those columns exist.
- **Publish:** click Publish to send the report to Power BI Service and share it. Optionally set a scheduled refresh.

Check your numbers: The Enrollments card should equal the clean_rows value in gold_data_quality (1,241), and Completed + In Progress + Not Started must add up to the same number.

13. Expected Deliverables

- A pipeline with Bronze, Silver (clean + rejected) and Gold tables.
- A short list of your cleaning rules and why you chose them.
- The gold_data_quality result showing the reconciliation check passed and the learner count before and after the email fix.
- One dashboard with three pages (Databricks AI/BI or Power BI).
- A README explaining how to set up and run the project.
- Screenshots of the pipeline graph, the Gold tables and the dashboard.

14. Acceptance Criteria / Definition of Done

- The CSV loads from the Volume and Bronze has 1,283 rows.
- Silver clean has no duplicate enrollments, no extra spaces, lower-case emails, no blank category or price, and no progress outside 0–100.
- Every clean enrollment has a learner_status of Completed, In Progress or Not Started.
- Rejected enrollments are saved with a reason.
- The four Gold business tables answer BR-01 to BR-04.
- gold_data_quality shows that bronze rows = clean + rejected + duplicates.
- The pipeline runs again with one click and the dashboard numbers match the Gold tables.

15. Questions for Project Review

1. Why do we keep the raw data in Bronze instead of cleaning the file directly?
2. Why do we use LOWER for emails but INITCAP for category names?
3. How many learners were double-counted before the email fix, and why does it matter?
4. Why can enrollment_id find duplicates but learner_email cannot?
5. Why are progress values of 0 and 100 valid, but 120 is rejected?
6. Why do we leave a blank rating as NULL instead of replacing it with 0?
7. Which category earns the most revenue, and which has the best completion rate?
8. Which dashboard tool did you choose, and why?

16. Final Business Outcome

At the end of the project, the learning platform has a repeatable pipeline that turns a messy enrollments file into clean, trusted summary tables and a simple dashboard. You can explain where the data came from, what problems you found, how you fixed them, why clean emails change the learner count, and how you proved the numbers are correct.

Key takeaway Medallion Architecture means each layer makes the data a little more trustworthy: Bronze keeps it, Silver cleans it, Gold summarises it, and the dashboard shares it.